PROGRAM 4: Generate Multiple Images Using Different Prompts

AIM

To generate multiple images using different text prompts and compare the effect of prompt variations.

ALGORITHM

1. Import the required libraries.
2. Load the pretrained text-to-image model.
3. Define different engineering-related prompts.
4. Generate an image for each prompt.
5. Display and compare the generated images.

CODE

```
import torch
from diffusers import AutoPipelineForText2Image
from IPython.display import display

if not torch.cuda.is_available():
    raise RuntimeError("Please enable GPU in Colab settings.")

pipe = AutoPipelineForText2Image.from_pretrained(
    "stabilityai/sd-turbo",
    torch_dtype=torch.float16,
    variant="fp16"
).to("cuda")

prompts = [
    "A small robot in a laboratory",
    "A futuristic humanoid robot in a laboratory",
    "A realistic industrial robot working in a factory"
]

for i, prompt in enumerate(prompts):
```

```
    print("\nPrompt:", prompt)

    image = pipe(
        prompt=prompt,
        num_inference_steps=1,
        guidance_scale=0.0,
        height=512,
        width=512
    ).images[0]

    display(image)

    image.save(f"image_{i+1}.png")

print("\nAll images generated successfully.")
```

INPUT

Three different prompts:

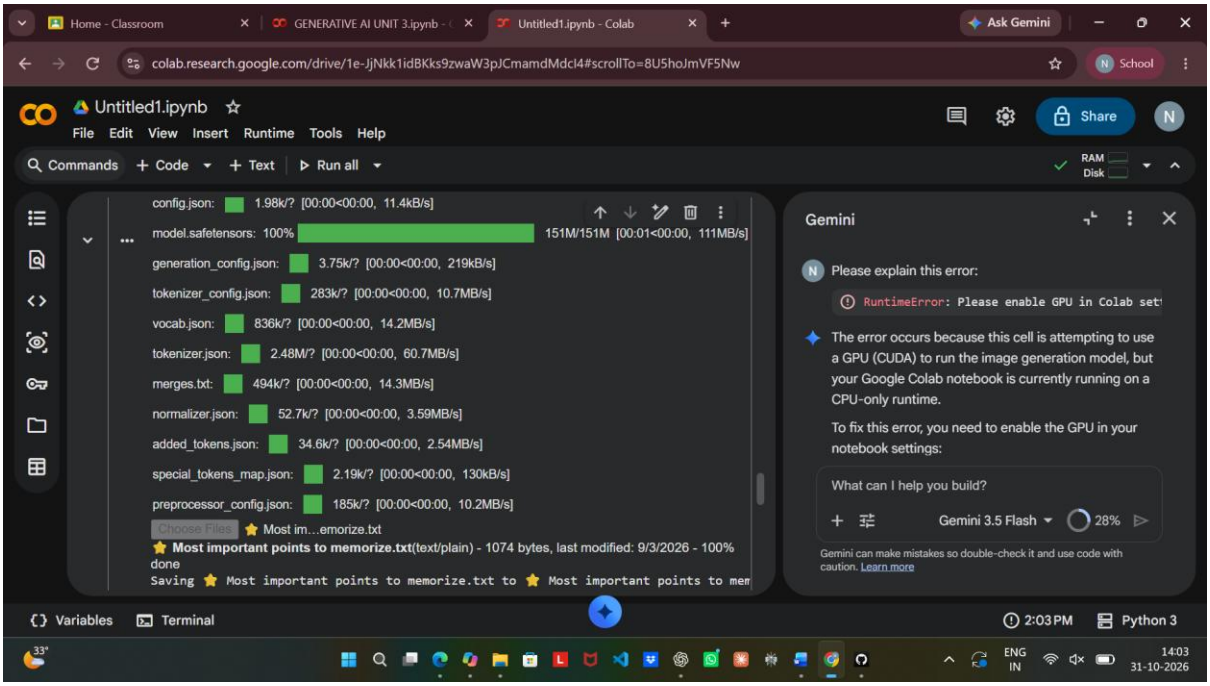
1. A small robot in a laboratory.
2. A futuristic humanoid robot in a laboratory.
3. A realistic industrial robot working in a factory.

OUTPUT

Three different AI-generated images are displayed.

Generated files:

- image_1.png
- image_2.png
- image_3.png

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RESULT

Thus, multiple images were generated successfully, and the effect of different text prompts was compared.